

Fabulus smoothing: main (MeshLib) vs GeometryEngine

Head-to-head over eleven clinical and synthetic inputs at the UI's standard preset · 27 September 2026

Introduction

Fabulus on main smooths a bolus with MeshLib voxel offsets and a geometry3Sharp Reducer. The geometry-engine branch smooths it with GeometryEngine's own DoubleOffset, Offset and Decimate. The two had never been run on the same input and compared, so this report does that: the same eleven meshes, the same settings, the same frame.

The question is not which is prettier, but whether the engine can replace main without changing the bolus a clinic prints — and, where it does change it, by how much and why.

How this was tested

Two headless harnesses generate the meshes, one per pipeline, each centring its input on the bounding box exactly as its own app does on import. The reference harness carries verbatim copies of main's smoothing code, since Fabulus.Core on main will not build off Windows.

Both run the UI's standard preset: a 1.0 mm offset cycle, 0.1 mm inflation, one iteration, 1.0 mm cell, decimated to twice the input triangle count. Component separation, which only the engine branch does, is off so both sides see the same geometry.

Measured with Fabulus.MeshCompare on linux-x64, against libmanifoldc built at upstream 7c86359 — the commit the shipped win-x64 binaries come from. ICP between the outputs returned under 0.04 mm and 0.18°, so the harnesses agree about where the model is.

Summary

The geometry matches closely: mean Dice 0.992, HD95 0.03 to 0.23 mm, mean surface separation under 0.07 mm. The engine sits a touch outside main, by +0.010 mm on average and never past +0.030 mm, which the code predicts.

Against the unsmoothed anatomy the engine is the closer of the two, halving main's HD95 on nine of eleven.

Topology was the gap and is now closed. Three outputs came out non-watertight, from the weld inside Decimate fusing a pinch the level-set offset legitimately emitted — not from the offset, whose guarantee held everywhere. All eleven are now watertight and manifold, Dice unchanged to four decimals.

What is left is self-intersections where the offset surface touches itself. Decimate no longer collapses across a contact while it has anywhere else to collapse, taking larynx small from ten to sixteen crossings down to none to four; chin_bolus keeps two or three. Read those counts with care — the offset is not reproducible run to run.

Per-case result

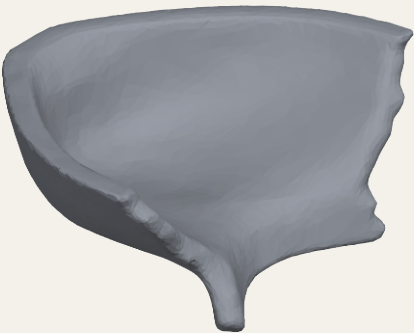
distances in mm; volume and area are the engine's difference from main; W / NM / SI = watertight, non-manifold edges, self-intersections

case	in tris	out tris	Dice	HD95	ASSD	bias	vol %	area %	main W / NM / SI	engine W / NM / SI	main s	engine s	checks
chin_bolus	3,216	6,432	0.9907	0.133	0.045	+0.007	+0.31	+1.17	yes / 0 / 0	yes / 0 / 3	1.52	3.45	self-intersections
ear_bolus	2,810	5,624	0.9898	0.183	0.047	+0.012	+0.53	+1.72	yes / 0 / 0	yes / 0 / 0	1.02	1.98	all pass
eye_bolus	1,154	2,308	0.9861	0.226	0.061	+0.029	+1.34	+2.64	yes / 0 / 0	yes / 0 / 0	0.63	0.91	volume, ICP noise
larynx small	2,978	5,960	0.9844	0.217	0.069	+0.016	+0.74	+2.44	yes / 0 / 0	yes / 0 / 4	0.68	2.27	topology, self-intersections
larynx_bolus	10,694	21,388	0.9961	0.121	0.032	+0.001	+0.04	+0.63	yes / 0 / 0	yes / 0 / 0	1.43	2.49	all pass
nose_bolus	1,566	3,132	0.9857	0.200	0.067	+0.012	+0.51	+2.34	yes / 0 / 0	yes / 0 / 0	0.67	0.99	all pass
scalp_bolus	8,546	17,092	0.9946	0.086	0.027	+0.004	+0.16	+0.59	yes / 0 / 0	yes / 0 / 0	1.29	3.25	all pass
small test	23,552	47,104	0.9988	0.037	0.022	+0.022	+0.24	+0.19	yes / 0 / 0	yes / 0 / 2	0.69	1.66	self-intersections, ICP noise
sphere	9,024	18,048	0.9997	0.034	0.016	+0.015	+0.06	+0.04	yes / 0 / 0	yes / 0 / 0	1.57	2.87	all pass
test bolus 107mL	15,212	30,424	0.9960	0.110	0.038	+0.007	+0.15	+0.69	yes / 0 / 0	yes / 0 / 0	0.92	2.68	all pass
test bolus 7mm	10,594	21,188	0.9947	0.133	0.039	-0.012	-0.32	+0.29	yes / 0 / 0	yes / 0 / 0	1.15	1.83	all pass

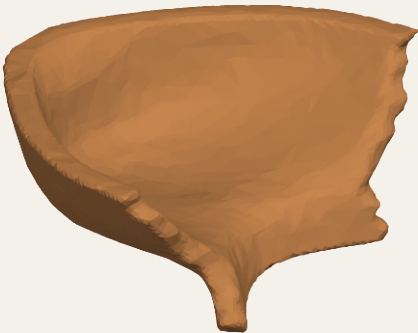
chin_bolus

3,216 input facets · 6,432 facets out of both pipelines · input surface 13,481 mm², volume 30,805 mm³

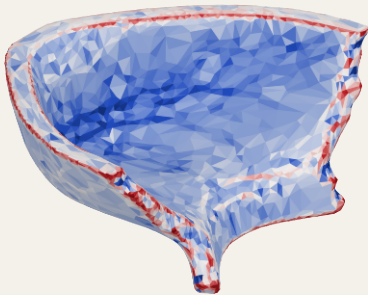
main (MeshLib)



GeometryEngine

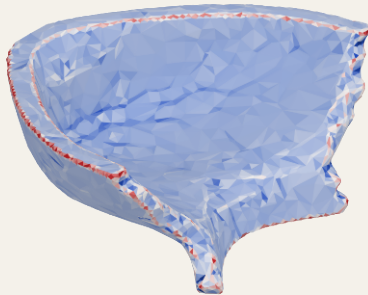


main vs original



mean |dev| 0.118 mm 34% inside

engine vs original



mean |dev| 0.097 mm 19% inside

The smoothed mesh from each pipeline. Every panel on this page shares one camera, fitted to the unsmoothed input.

Metrics

engine vs main

Dice	0.9907
HD95 / Hausdorff	0.133 / 0.806 mm
mean separation (ASSD)	0.0451 mm
signed bias	+0.0072 mm
surface Dice @0.5 mm	0.9999
volume / area	+0.31 % / +1.17 %
curvature ratio	1.236

topology

	main	engine
watertight	yes	yes
non-manifold edges	0	0
self-intersections	0	3
components	1	1
genus	0	0
degenerate facets	0	2

fidelity to the input

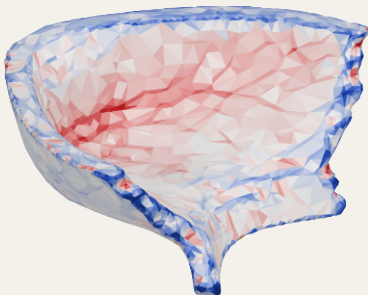
Dice	0.9773 → 0.9788
HD95	0.253 → 0.147 mm
mean separation	0.108 → 0.100 mm
signed bias	+0.083 → +0.090 mm
volume	+3.62 → +3.94 %
runtime	1.52 → 3.45 s
ICP residual	0.032 mm, 0.052°

main → engine

meshcompare default checks

- `cand.self_intersections==0` (actual 3)
- `cand.degenerate_triangles==0` (actual 2)

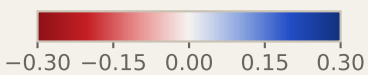
engine vs main



mean |dev| 0.068 mm 33% inside

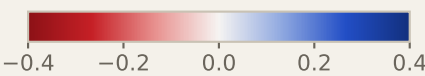
Signed deviation

panel above · ±0.3 mm



Signed deviation

both panels above · ±0.4 mm



Each panel colours the mesh named first in its title by how far it lies from the mesh named second. Red: the coloured surface lies inside the one it is measured against — material the smoothing took away. Blue: outside it — material the smoothing added. Near-white: the two surfaces coincide.

The two fidelity panels share one scale so they can be read against each other. The head-to-head below has its own, because the pipelines differ from each other by far less than either differs from the input.

ear_bolus

2,810 input facets · 5,624 facets out of both pipelines · input surface 11,134 mm², volume 23,914 mm³

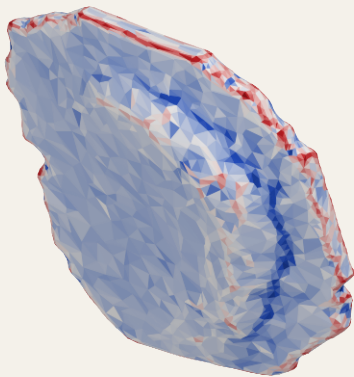
main (MeshLib)



GeometryEngine

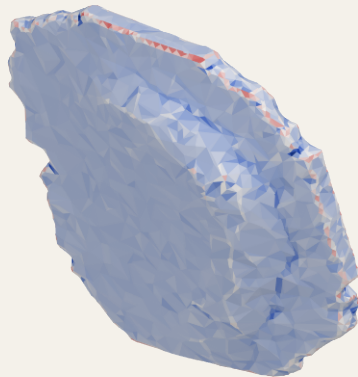


main vs original



mean |dev| 0.127 mm 34% inside

engine vs original



mean |dev| 0.095 mm 15% inside

The smoothed mesh from each pipeline. Every panel on this page shares one camera, fitted to the unsmoothed input.

Metrics

engine vs main

Dice	0.9898
HD95 / Hausdorff	0.183 / 0.697 mm
mean separation (ASSD)	0.0467 mm
signed bias	+0.0121 mm
surface Dice @0.5 mm	0.9999
volume / area	+0.53 % / +1.72 %
curvature ratio	1.284

topology

	main	engine
watertight	yes	yes
non-manifold edges	0	0
self-intersections	0	0
components	1	1
genus	0	0
degenerate facets	0	0

fidelity to the input

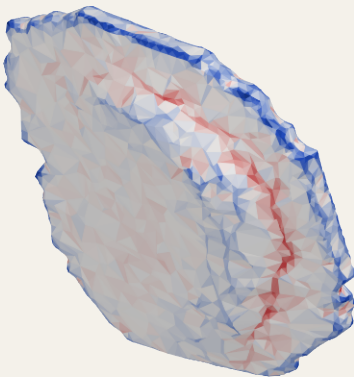
Dice	0.9747 → 0.9776
HD95	0.273 → 0.145 mm
mean separation	0.113 → 0.099 mm
signed bias	+0.082 → +0.094 mm
volume	+3.80 → +4.35 %
runtime	1.02 → 1.98 s
ICP residual	0.008 mm, 0.014°

main → engine

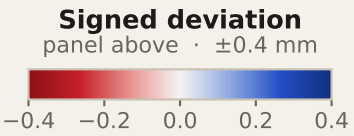
meshcompare default checks

all ten pass

engine vs main

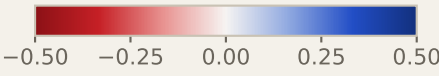


mean |dev| 0.084 mm 35% inside



Signed deviation

both panels above · ±0.5 mm



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eye_bolus

1,154 input facets · 2,308 facets out of both pipelines · input surface 4,760 mm², volume 9,847 mm³

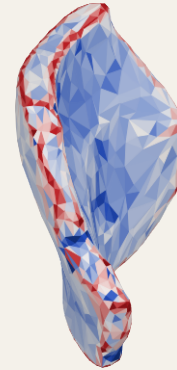
main (MeshLib)



GeometryEngine

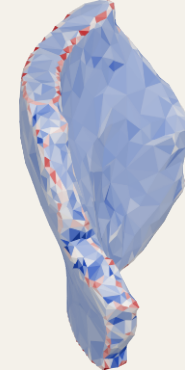


main vs original



mean |dev| 0.122 mm 43% inside

engine vs original



mean |dev| 0.089 mm 21% inside

The smoothed mesh from each pipeline. Every panel on this page shares one camera, fitted to the unsmoothed input.

Metrics

engine vs main

Dice	0.9861
HD95 / Hausdorff	0.226 / 0.814 mm
mean separation (ASSD)	0.0612 mm
signed bias	+0.0295 mm
surface Dice @0.5 mm	0.9997
volume / area	+1.34 % / +2.64 %
curvature ratio	1.270

fidelity to the input

Dice	0.9738 → 0.9773
HD95	0.284 → 0.146 mm
mean separation	0.112 → 0.096 mm
signed bias	+0.059 → +0.087 mm
volume	+2.84 → +4.21 %
runtime	0.63 → 0.91 s
ICP residual	0.018 mm, 0.102°

main → engine

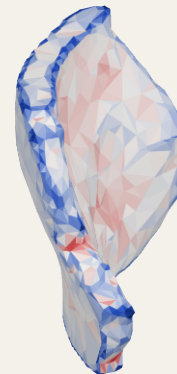
topology

	main	engine
watertight	yes	yes
non-manifold edges	0	0
self-intersections	0	0
components	1	1
genus	0	0
degenerate facets	0	0

meshcompare default checks

- cand_ref.abs_volume_diff_pct ≤ 1 (actual 1.33772)
- icp.rotation_deg ≤ 0.1 (actual 0.101575)

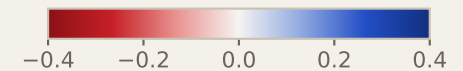
engine vs main



mean |dev| 0.102 mm 24% inside

Signed deviation

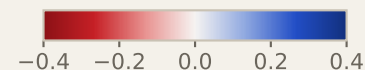
both panels above · ±0.4 mm



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Signed deviation

panel above · ±0.4 mm

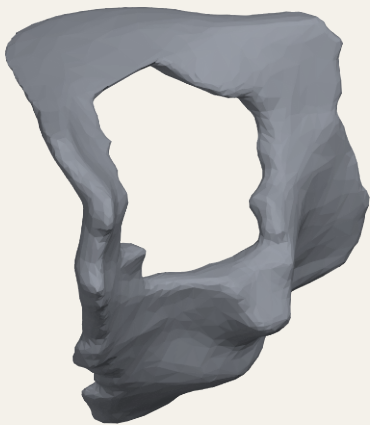


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larynx small

2,978 input facets · 5,960 facets out of both pipelines · input surface 14,058 mm², volume 28,494 mm³

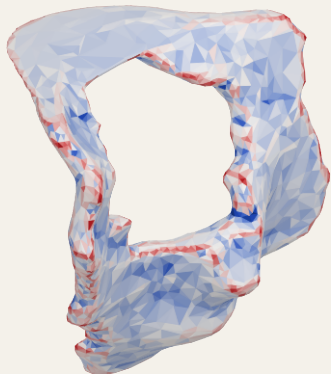
main (MeshLib)



GeometryEngine

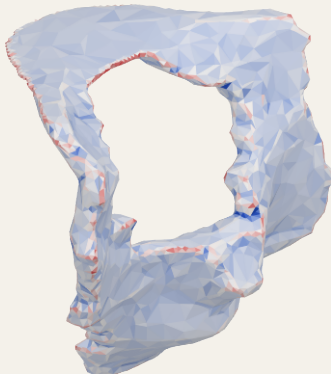


main vs original



mean |dev| 0.145 mm 40% inside

engine vs original



mean |dev| 0.106 mm 25% inside

The smoothed mesh from each pipeline. Every panel on this page shares one camera, fitted to the unsmoothed input.

Metrics

engine vs main

Dice	0.9844
HD95 / Hausdorff	0.217 / 1.320 mm
mean separation (ASSD)	0.0690 mm
signed bias	+0.0164 mm
surface Dice @0.5 mm	0.9983
volume / area	+0.74 % / +2.44 %
curvature ratio	1.227

topology

	main	engine
watertight	yes	yes
non-manifold edges	0	0
self-intersections	0	4
components	1	1
genus	1	2
degenerate facets	0	0

fidelity to the input

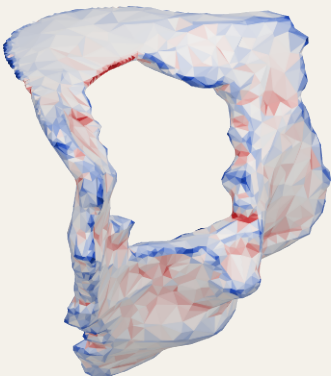
Dice	0.9705 → 0.9758
HD95	0.398 → 0.220 mm
mean separation	0.130 → 0.105 mm
signed bias	+0.064 → +0.080 mm
volume	+3.20 → +3.97 %
runtime	0.68 → 2.27 s
ICP residual	0.007 mm, 0.018°

main → engine

meshcompare default checks

- `cand.self_intersections==0` (actual 4)
- `cand.genus==ref.genus` (actual 2)

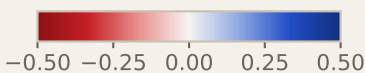
engine vs main



mean |dev| 0.102 mm 35% inside

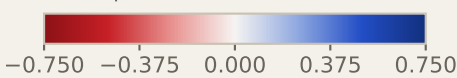
Signed deviation

panel above · ±0.5 mm



Signed deviation

both panels above · ±0.75 mm



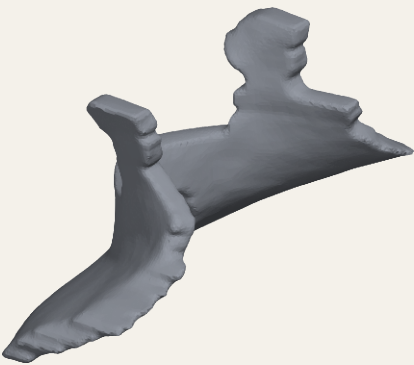
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larynx_bolus

10,694 input facets · 21,388 facets out of both pipelines · input surface 45,689 mm², volume 178,457 mm³

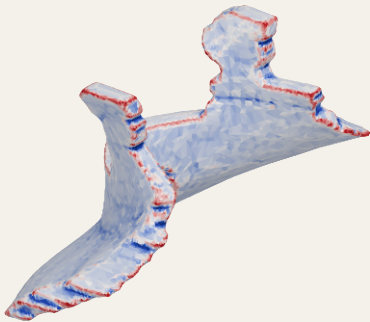
main (MeshLib)



GeometryEngine

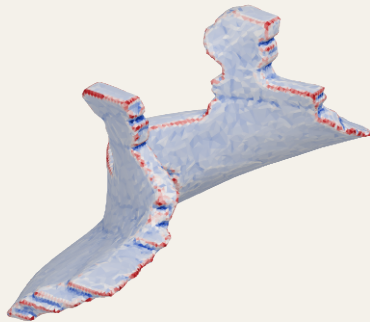


main vs original



mean |dev| 0.141 mm 31% inside

engine vs original



mean |dev| 0.119 mm 23% inside

The smoothed mesh from each pipeline. Every panel on this page shares one camera, fitted to the unsmoothed input.

Metrics

engine vs main

Dice	0.9961
HD95 / Hausdorff	0.121 / 1.228 mm
mean separation (ASSD)	0.0322 mm
signed bias	+0.0015 mm
surface Dice @0.5 mm	0.9990
volume / area	+0.04 % / +0.63 %
curvature ratio	1.030

topology

	main	engine
watertight	yes	yes
non-manifold edges	0	0
self-intersections	0	0
components	1	1
genus	1	1
degenerate facets	0	0

fidelity to the input

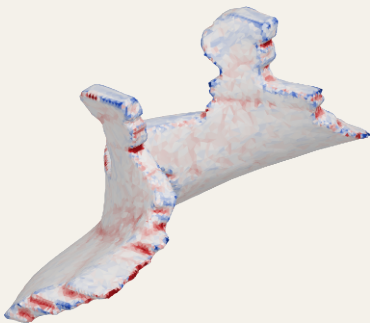
Dice	0.9850 → 0.9863
HD95	0.364 → 0.270 mm
mean separation	0.123 → 0.112 mm
signed bias	+0.070 → +0.072 mm
volume	+1.82 → +1.85 %
runtime	1.43 → 2.49 s
ICP residual	0.004 mm, 0.002°

main → engine

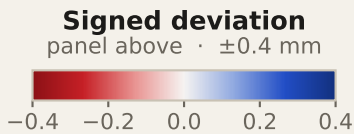
meshcompare default checks

all ten pass

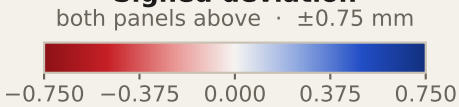
engine vs main



mean |dev| 0.049 mm 47% inside



Signed deviation



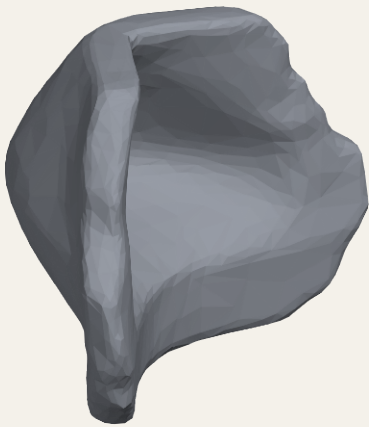
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nose_bolus

1,566 input facets · 3,132 facets out of both pipelines · input surface 6,030 mm², volume 13,294 mm³

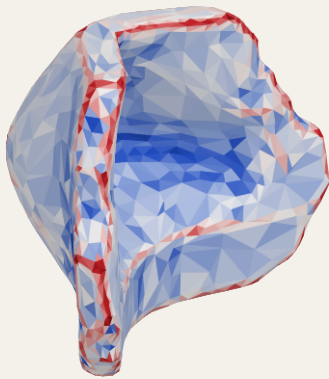
main (MeshLib)



GeometryEngine

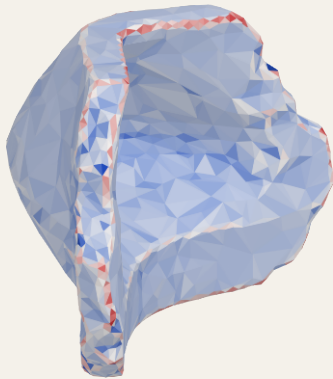


main vs original



mean |dev| 0.134 mm 37% inside

engine vs original



mean |dev| 0.096 mm 17% inside

The smoothed mesh from each pipeline. Every panel on this page shares one camera, fitted to the unsmoothed input.

Metrics

engine vs main

Dice	0.9857
HD95 / Hausdorff	0.200 / 0.639 mm
mean separation (ASSD)	0.0669 mm
signed bias	+0.0119 mm
surface Dice @0.5 mm	0.9999
volume / area	+0.51 % / +2.34 %
curvature ratio	1.228

topology

	main	engine
watertight	yes	yes
non-manifold edges	0	0
self-intersections	0	0
components	1	1
genus	0	0
degenerate facets	0	0

fidelity to the input

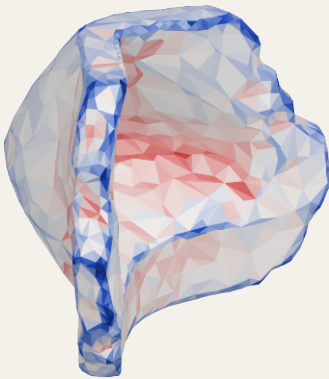
Dice	0.9739 → 0.9783
HD95	0.336 → 0.174 mm
mean separation	0.121 → 0.099 mm
signed bias	+0.080 → +0.091 mm
volume	+3.59 → +4.12 %
runtime	0.67 → 0.99 s
ICP residual	0.030 mm, 0.092°

main → engine

meshcompare default checks

all ten pass

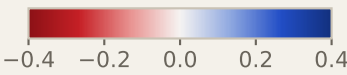
engine vs main



mean |dev| 0.091 mm 35% inside

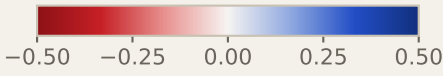
Signed deviation

panel above · ±0.4 mm



Signed deviation

both panels above · ±0.5 mm



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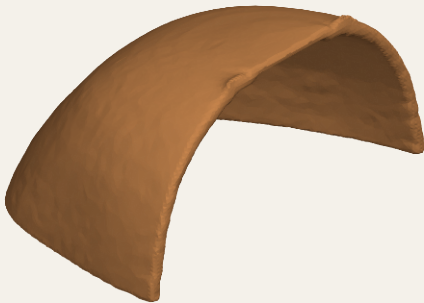
scalp_bolus

8,546 input facets · 17,092 facets out of both pipelines · input surface 42,347 mm², volume 100,832 mm³

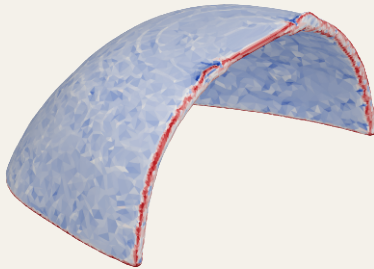
main (MeshLib)



GeometryEngine

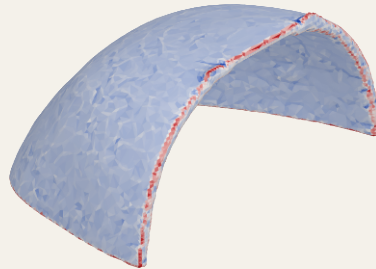


main vs original



mean |dev| 0.120 mm 26% inside

engine vs original



mean |dev| 0.098 mm 14% inside

The smoothed mesh from each pipeline. Every panel on this page shares one camera, fitted to the unsmoothed input.

Metrics

engine vs main

Dice	0.9946
HD95 / Hausdorff	0.086 / 0.658 mm
mean separation (ASSD)	0.0270 mm
signed bias	+0.0040 mm
surface Dice @0.5 mm	1.0000
volume / area	+0.16 % / +0.59 %
curvature ratio	1.058

topology

	main	engine
watertight	yes	yes
non-manifold edges	0	0
self-intersections	0	0
components	1	1
genus	0	0
degenerate facets	0	0

fidelity to the input

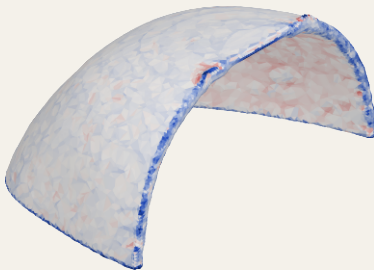
Dice	0.9778 → 0.9796
HD95	0.193 → 0.138 mm
mean separation	0.109 → 0.100 mm
signed bias	+0.085 → +0.089 mm
volume	+3.58 → +3.74 %
runtime	1.29 → 3.25 s
ICP residual	0.037 mm, 0.023°

main → engine

meshcompare default checks

all ten pass

engine vs main



mean |dev| 0.045 mm 40% inside

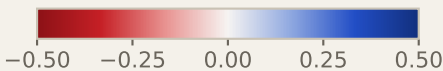
Signed deviation

panel above · ±0.3 mm



Signed deviation

both panels above · ±0.5 mm



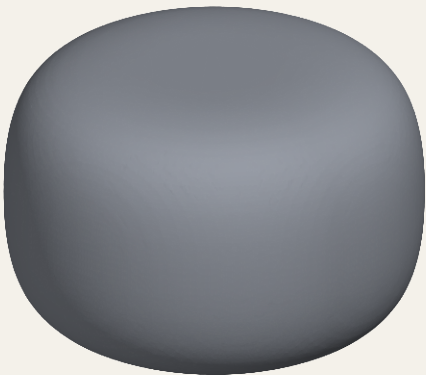
Each panel colours the mesh named first in its title by how far it lies from the mesh named second. Red: the coloured surface lies inside the one it is measured against — material the smoothing took away. Blue: outside it — material the smoothing added. Near-white: the two surfaces coincide.

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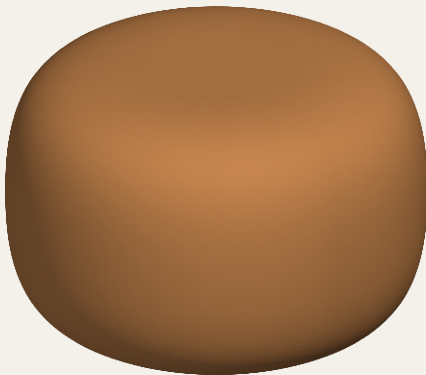
small test

23,552 input facets · 47,104 facets out of both pipelines · input surface 10,575 mm², volume 96,627 mm³

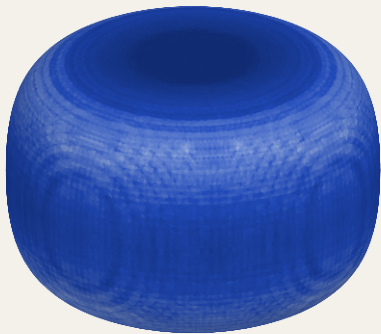
main (MeshLib)



GeometryEngine

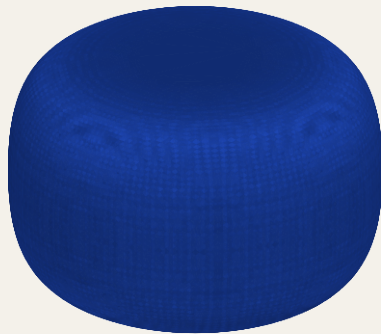


main vs original



mean |dev| 0.062 mm 0% inside

engine vs original



mean |dev| 0.089 mm 0% inside

The smoothed mesh from each pipeline. Every panel on this page shares one camera, fitted to the unsmoothed input.

Metrics

engine vs main

Dice	0.9988
HD95 / Hausdorff	0.037 / 0.056 mm
mean separation (ASSD)	0.0219 mm
signed bias	+0.0219 mm
surface Dice @0.5 mm	1.0000
volume / area	+0.24 % / +0.19 %
curvature ratio	0.810

topology

	main	engine
watertight	yes	yes
non-manifold edges	0	0
self-intersections	0	2
components	1	1
genus	0	0
degenerate facets	0	0

fidelity to the input

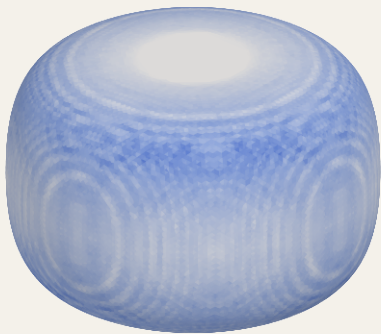
Dice	0.9963 → 0.9951
HD95	0.099 → 0.100 mm
mean separation	0.069 → 0.090 mm
signed bias	+0.069 → +0.090 mm
volume	+0.75 → +0.99 %
runtime	0.69 → 1.66 s
ICP residual	0.001 mm, 0.178°

main → engine

meshcompare default checks

- `cand.self_intersections==0` (actual 2)
- `icp.rotation_deg<=0.1` (actual 0.17754)

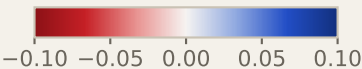
engine vs main



mean |dev| 0.023 mm 0% inside

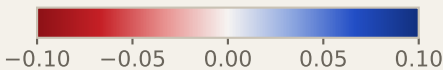
Signed deviation

panel above · ±0.1 mm



Signed deviation

both panels above · ±0.1 mm

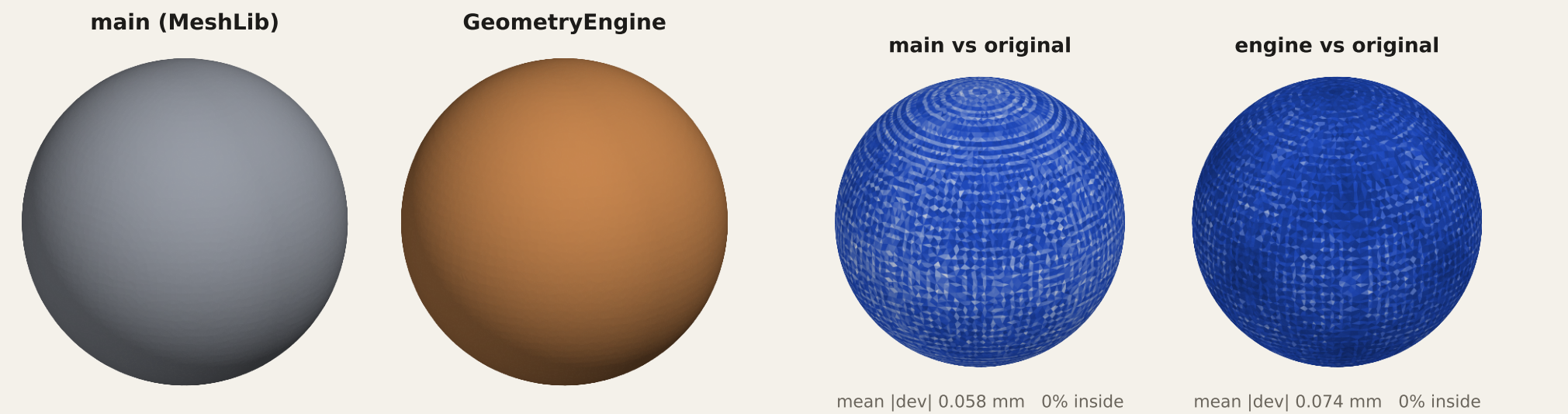


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sphere

9,024 input facets · 18,048 facets out of both pipelines · input surface 80,353 mm², volume 2,140,815 mm³

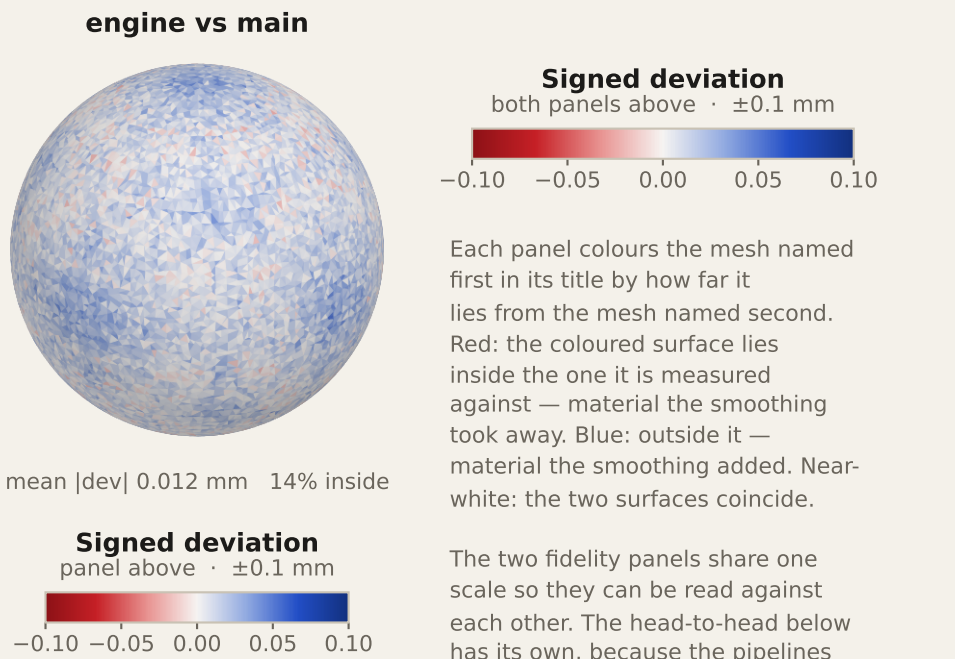


The smoothed mesh from each pipeline. Every panel on this page shares one camera, fitted to the unsmoothed input.

Metrics

engine vs main		fidelity to the input		main → engine	
Dice	0.9997	Dice	0.9988 → 0.9985		
HD95 / Hausdorff	0.034 / 0.081 mm	HD95	0.084 → 0.096 mm		
mean separation (ASSD)	0.0161 mm	mean separation	0.064 → 0.079 mm		
signed bias	+0.0154 mm	signed bias	+0.064 → +0.079 mm		
surface Dice @0.5 mm	1.0000	volume	+0.24 → +0.30 %		
volume / area	+0.06 % / +0.04 %	runtime	1.57 → 2.87 s		
curvature ratio	0.963	ICP residual	0.001 mm, 0.076°		

topology	main	engine	meshcompare default checks
watertight	yes	yes	all ten pass
non-manifold edges	0	0	
self-intersections	0	0	
components	1	1	
genus	0	0	
degenerate facets	0	0	

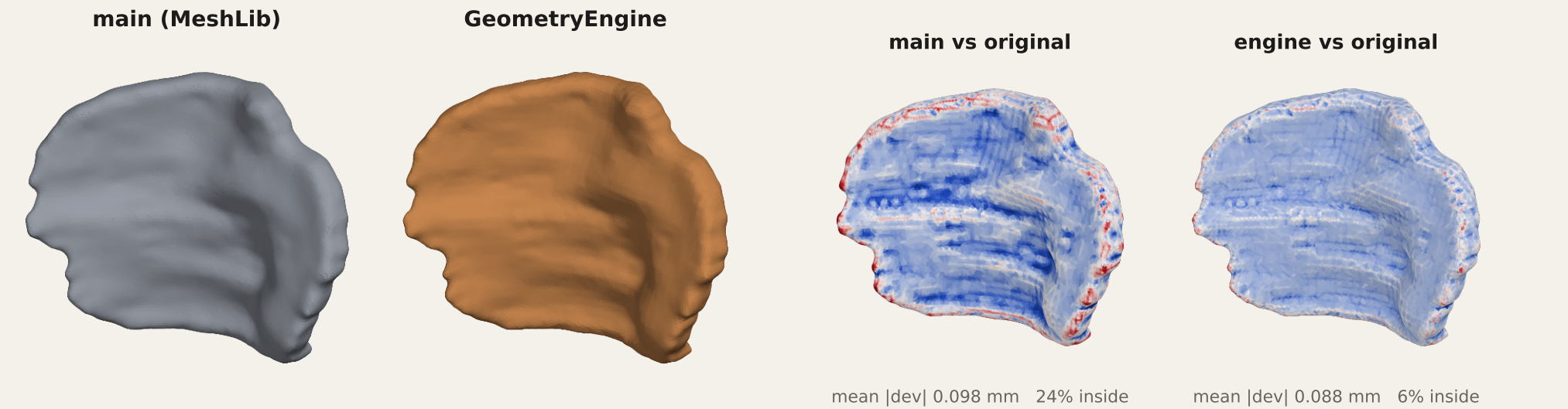


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test bolus 107mL

15,212 input facets · 30,424 facets out of both pipelines · input surface 23,295 mm², volume 106,883 mm³



The smoothed mesh from each pipeline. Every panel on this page shares one camera, fitted to the unsmoothed input.

Metrics

engine vs main

Dice	0.9960
HD95 / Hausdorff	0.110 / 0.599 mm
mean separation (ASSD)	0.0379 mm
signed bias	+0.0072 mm
surface Dice @0.5 mm	1.0000
volume / area	+0.15 % / +0.69 %
curvature ratio	0.964

fidelity to the input

Dice	0.9893 → 0.9900
HD95	0.224 → 0.144 mm
mean separation	0.100 → 0.093 mm
signed bias	+0.083 → +0.090 mm
volume	+1.80 → +1.96 %
runtime	0.92 → 2.68 s
ICP residual	0.012 mm, 0.008°

main → engine

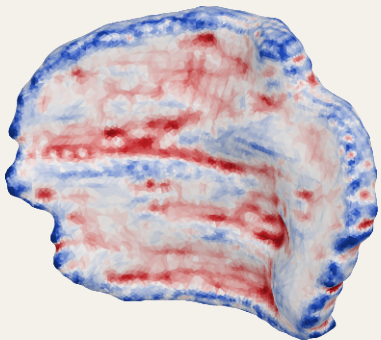
topology

	main	engine
watertight	yes	yes
non-manifold edges	0	0
self-intersections	0	0
components	1	1
genus	0	0
degenerate facets	0	0

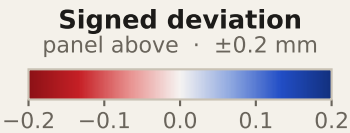
meshcompare default checks

all ten pass

engine vs main



mean |dev| 0.045 mm 37% inside



Signed deviation

both panels above · ±0.4 mm

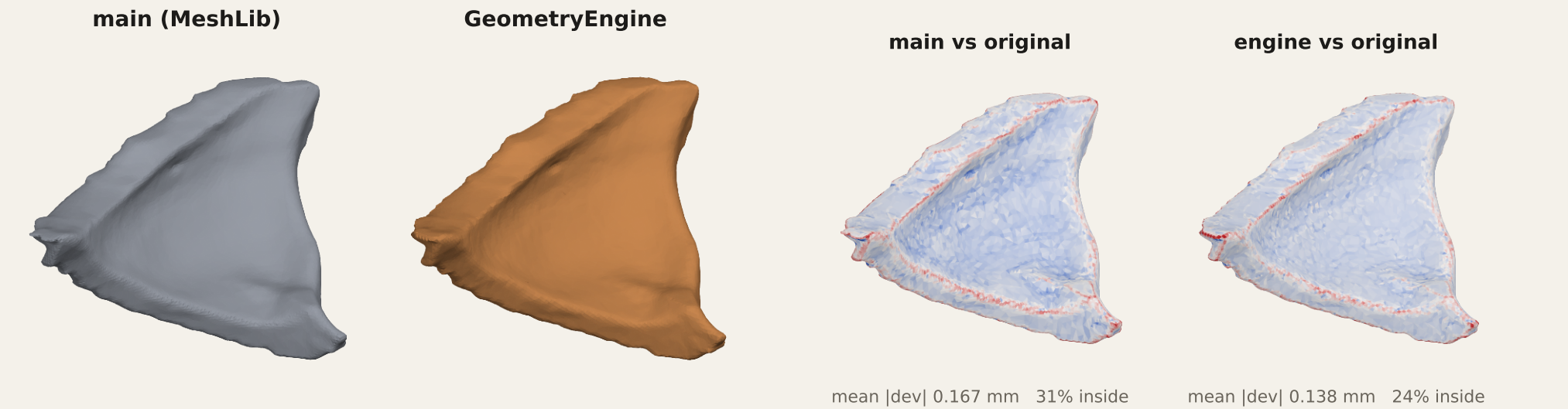


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test bolus 7mm

10,594 input facets · 21,188 facets out of both pipelines · input surface 56,349 mm², volume 191,158 mm³



The smoothed mesh from each pipeline. Every panel on this page shares one camera, fitted to the unsmoothed input.

Metrics

engine vs main

Dice	0.9947
HD95 / Hausdorff	0.133 / 3.425 mm
mean separation (ASSD)	0.0387 mm
signed bias	-0.0121 mm
surface Dice @0.5 mm	0.9987
volume / area	-0.32 % / +0.29 %
curvature ratio	0.920

fidelity to the input

main → engine

Dice	0.9803 → 0.9825
HD95	0.478 → 0.393 mm
mean separation	0.141 → 0.128 mm
signed bias	+0.088 → +0.074 mm
volume	+2.61 → +2.29 %
runtime	1.15 → 1.83 s
ICP residual	0.013 mm, 0.005°

topology

main

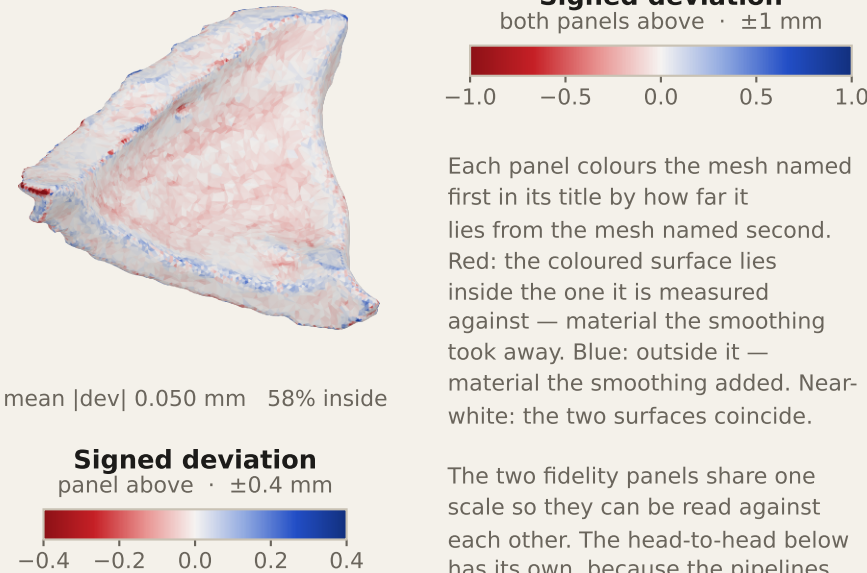
engine

watertight	yes	yes
non-manifold edges	0	0
self-intersections	0	0
components	1	1
genus	0	0
degenerate facets	0	0

meshcompare default checks

all ten pass

engine vs main



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